

Logical Operators Quick Reference				
$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$
$\forall$	$\exists$	$\in$	$\notin$	$\equiv$

1. Let the following variables represent the corresponding phrases:

p – “Today is Friday”

q – “Papa Johns is on fire”

r – “win a chess match”

s – “watch a movie”

t – “will sleep early”

u – “will get to class on time”

v – “Turn in my homework”

w – “Dominoes is on fire today”

1	Today is Friday and Papa Johns is on fire.	$p \wedge q$
2	I will win a chess match if today is a Friday or I do not watch a movie.	$(p \vee \neg s) \rightarrow r$
3	I sleep early only if I do not win a chess match.	$t \rightarrow \neg r$
4	I will get to class on time and turn in my homework if I do not sleep early.	$\neg t \rightarrow (u \wedge v)$
5	Papa Johns is not on fire if and only if I do not turn in my homework.	$\neg q \leftrightarrow \neg v$
6	Dominoes is not on fire today.	$\neg w$

Where the conclusion is: $\neg(v \rightarrow t)$

7	p	using (1) by Simplification
7A	$p \vee \neg s$	using (2) by Addition
8	$p \rightarrow r$	using (2) and (7) by Identity Laws
9	$\neg p \vee r$	using (8) by Conjunctive Disjunction
10	$\neg t \vee \neg r$	using (3) by Conjunctive Disjunction
11	$\neg p \vee \neg t$	using (9) and (10) by Resolution
12	$\neg t$	using (7) and (11) by Identity Laws
13	$t \vee (u \wedge v)$	using (4) by Conjunctive Disjunction
14	$(u \wedge v)$	using (12) and (13) by Identity Laws
15	v	using (14) by Simplification
16	$v \wedge \neg t$	using (12) and (15) by Conjunction
17	$\neg(\neg v \vee t)$	using (16) by DeMorgan's Law
18	$\neg(v \rightarrow t)$	using (17) by Conjunctive Disjunction

- 2.
- (1) $y \leftrightarrow x$
 - (2) $x \wedge (b \vee \neg d)$
 - (3) $x \wedge a \rightarrow \neg b$
 - (4) $(\neg y \vee x) \wedge c \rightarrow d$
 - (5) $y \rightarrow c$

b

6	x	using (2) by Simplification
7	$(y \rightarrow x) \wedge (x \rightarrow y)$	using (1) by Definition of Biconditional
8	$(y \rightarrow x) \wedge y$	using (6) and (7) by Modus Ponens
9	y	using (8) by Simplification
10	c	using (5) and (9) by Modus Ponens
11	$x \wedge c \rightarrow d$	using (4) and (6) by Addition
12	$c \rightarrow d$	using (11) by Simplification
13	d	using (10) and (12) by Modus Ponens
14	$b \vee \neg d$	using (2) by Simplification
15	b	using (13) and (14) by Disjunctive Syllogism

3. Let x and y be integers. Show that if $x + y$ is even, then $x^2y - y^3 + 2$ is even.

We will proceed by a direct proof. Let $x + y = z$ where z is an even integer. By definition of even numbers, $z = 2k$ where k is some integer.

We will show that if $x + y = z$ is even, then $x - y = z_0$ is also even.

$x + y = 2k$	Substitution
$y = 2k - x$	Subtracting x on both sides
$x - y = z_0$	Given
$x - (2k - x) = z_0$	Substitution
$2x - 2k = z_0$	Expansion of parenthesis
$2(x - k) = z_0$	Factoring

We know $x - k \in \mathbb{Z}$ since integers are closed under subtraction. Therefore, by definition of even numbers, $x - y$ is even.

Using definition of even numbers and established relationships, we will proceed with the rest of the proof.

$x^2y - y^3 + 2$	Given
$y(x^2 - y^2) + 2$	Factoring out y
$y(x + y)(x - y) + 2$	Factoring out perfect squares
$(2k - x)(2x - 2k)(2k) + 2$	Substitution
$(4kx - 4k^2 - 2x^2 + 2kx)(2k) + 2$	Distribution
$8k^2x - 8k^3 - 4x^2k + 4k^2x + 2$	Distribution
$12k^2x - 8k^3 - 4x^2k + 2$	Combining like terms
$2(6k^2x - 4k^3 - 2x^2k + 1)$	Factoring out 2

We know $6k^2x - 4k^3 - 2x^2k + 1 \in \mathbb{Z}$ since integers are closed under addition, subtraction, and multiplication and from previously established connections. Therefore, by definition of even numbers, $x^2y - y^3 + 2$ is even and the proof is concluded. ■

4A. Let n be an integer. Prove that if $n^2 + 2n + 10$ is odd, then n is odd.

We will proceed by contraposition. Let $p(n) = \text{"is even"}$. We wish to prove $\forall n (p(n^2 + 2n + 10) \rightarrow p(n))$ where $n \in \mathbb{Z}$. The contrapositive of the conditional statement is $\neg p(n) \rightarrow \neg p(n^2 + 2n + 10)$, or equivalently, if n is even, then $n^2 + 2n + 10$ is even. Since n is even, then, by definition of even numbers, $n = 2k$ where k is an integer.

$n^2 + 2n + 10$	Given
$(2k)^2 + 2(2k) + 10$	Substitution
$4k^2 + 4k + 10$	Distribution
$2(2k^2 + 2k + 5)$	Factoring out 2

We know $2k^2 + 2k + 5 \in \mathbb{Z}$ since integers are closed under subtraction. Therefore, by definition of even numbers, $n^2 + 2n + 10$ is even. We conclude $n^2 + 2n + 10$ is even and finish our proof. ■

4B. Let n be an integer. Prove that if $n^2 + 2n + 10$ is odd, then n is odd.

We will proceed by contradiction. Let $p(n) = \text{"is even"}$. We wish to prove $\forall n (p(n^2 + 2n + 10) \rightarrow p(n))$ where $n \in \mathbb{Z}$. By contradiction, we wish to show $n^2 + 2n + 10$ is odd but n is even. Since n is even, by contradiction, by the definition of even numbers, $n = 2k$ where k is an integer.

$n^2 + 2n + 10$	Given
$(2k)^2 + 2(2k) + 10$	Substitution
$4k^2 + 4k + 10$	Distribution

$2(2k^2 + 2k + 5)$	Factoring out 2
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We know $2k^2 + 2k + 5 \in \mathbb{Z}$ since integers are closed under subtraction. Therefore, by definition of even numbers, $n^2 + 2n + 10$ is even. However, by the original statement, $n^2 + 2n + 10$ is odd, which is a contradiction. Then our assumption is false, and we conclude n is odd, as desired. ■

5. Prove that $\sqrt[n]{2}$ is irrational for $n \geq 3$.

We will proceed by contradiction. Assume $\sqrt[n]{2}$ is rational for $n \geq 3$. Let $a, b \in \mathbb{Z}$, where $b \neq 0$.

By definition of rational numbers, we may establish $\sqrt[n]{2} = \frac{a}{b}$ where a and b have no common factor.

$\sqrt[n]{2} = \frac{a}{b}$	Given
$b\sqrt[n]{2} = a$	Multiply b on both sides
$2b^n = a^n$	Raising each side to n

We know $b^n \in \mathbb{Z}$ since integers are closed under multiplication. Therefore, by definition of even numbers, we know a is even. Therefore, there exists an $a \in \mathbb{Z}$ such that $a = 2a$. Therefore, we have the following:

$2b^n = (2a)^n = 2^n a^n$	Substitution
$b^n = 2^{n-1} a^n$	Dividing each side by 2

We know $a^n \in \mathbb{Z}$ since integers are closed under multiplication. Therefore, by definition of even numbers, we know b is even. If a and b are even, then a and b share a factor of 2, contradicting our assumptions. Hence, $\sqrt[n]{2}$ must be irrational. The proof is complete. ■

- 6A. The most efficient method for proving the statement is proof by contraposition.

There are two distinct statements, $p(x) = "x^3 - x^2 + 1 \text{ is even}"$ and $q(x) = "the x\text{-th power of the } x\text{-th prime is always odd}"$. The format of these statements give way for a simple negation of both of them, allowing for the proof to show $\neg p(x) = "x^3 - x^2 + 1 \text{ is odd}"$ and $\neg q(x) = "the x\text{-th power and the } x\text{-th prime is always positive}"$. With regard to proving the proof, it would be sufficient to show if $\neg q(x)$, then $\neg p(x)$, thereby proving the proof.

- 6B. The most efficient method for proving the statement is proof by contradiction.

There are two distinct statements, $p(x) = \frac{x^4}{\log(x)} > (\ln x)^{1/3}$ and $q(x) = x^3 + x > x^2 - x$.

Due to the more simplistic nature of the second statement, $q(x)$, it is more reasonable to find a value of x such that x is a positive integer and agrees with $p(x)$ while also disagreeing with $q(x)$. This is possible due to the nature of logarithms and the properties held with logarithms and positive values, thereby aiding in the proof by making it convenient to show agreement with $p(x)$ and disagreement with $q(x)$ since positive numbers, behave similarly for logarithms.